

The flowed bound in free field theory: the coupling cancels, and the bound is informative only at the confinement scale

The two ground-state quantities of the finiteness note are computed in closed form for the free (abelian) theory with the flowed magnetic energy density $\varphi_t = \frac{1}{2}(b_t \cdot b_t - \langle \cdot \rangle)$:

$$\chi_{\varphi_t} = \frac{3g^4}{2048 \pi^{3/2} t^{5/2}}, \quad \langle |D_{\varphi_t}|^2 \rangle = \frac{g^2}{128 \pi^2 t^3}, \quad \frac{\langle |D|^2 \rangle}{\chi} = \frac{16}{3 g^2 \sqrt{\pi t}},$$

so that the bound $m \leq \frac{g^2 \hbar c}{2} \langle |D|^2 \rangle / \chi$ becomes

$$m \leq \frac{8}{3\sqrt{\pi}} \frac{\hbar c}{\sqrt{t}} = \frac{16\sqrt{2}}{3\sqrt{\pi}} \frac{\hbar c}{\sqrt{8t}} \approx 4.26 \frac{\hbar c}{\sqrt{8t}}$$

with the coupling cancelling between numerator and denominator. This corrects Section 5 of the finiteness note, which carried a factor g^2 from dimensional analysis alone: $\chi \propto g^4$ and $\langle |D|^2 \rangle \propto g^2$, so the ratio supplies g^{-2} . The result is confirmed independently by the spectral form of the same bound: the smeared operator creates two photons of momenta $\pm k$ with weight $\propto k^4 e^{-4tk^2} dk$, and the weighted mean of their energy $2\hbar ck$ is exactly $8\hbar c/(3\sqrt{\pi t})$. Two consequences. The free bound tends to zero as the smearing grows, as it must for $m = 0$, so the inequality retains content. And in the interacting theory the weak-coupling regime is $\sqrt{8t} \ll \hbar c/(\text{confinement scale})$, where the bound reads $m \lesssim 4.26 \hbar c/\sqrt{8t}$ and is far weaker than the expected gap: the bound becomes informative exactly at $\sqrt{8t} \sim \hbar c/m$, where its coefficient is no longer perturbative. No perturbative computation can make the finiteness half useful; what is needed is the value of one susceptibility at the confinement scale. Constants explicit; nothing promoted.

1. The free ground state with explicit constants

Take the abelian case of the Hamiltonian of the finiteness note §1,

$$H = \int d^3x \left[\frac{g^2 c}{2\hbar} \Pi_i \Pi_i + \frac{\hbar c}{2g^2} b_i b_i \right], \quad b_i = \varepsilon_{ijk} \partial_j A_k, \quad [A_i(x), \Pi_j(y)] = i\hbar \delta_{ij} \delta^3(x - y),$$

in Coulomb gauge $\partial_i A_i = 0$, two transverse polarizations. In Fourier modes each transverse mode is an oscillator of mass $M = \hbar/(g^2 c)$ and frequency $\omega_k = ck$, since $\frac{1}{2} M \omega_k^2 = \hbar ck^2/(2g^2)$ matches the magnetic term. Its ground-state width gives

$$\langle \hat{A}_i(k) \hat{A}_j(k') \rangle_0 = (2\pi)^3 \delta^3(k + k') P_{ij}^T(k) \frac{\hbar}{2M\omega_k} = (2\pi)^3 \delta^3(k + k') P_{ij}^T(k) \frac{g^2}{2k},$$

with $P_{ij}^T = \delta_{ij} - \hat{k}_i \hat{k}_j$. Dimensions: $[\hat{A}] = L^2$, $[(2\pi)^3 \delta^3] = L^3$, and $g^2/2k$ is a length.

2. The flow is the heat equation here

For an abelian field the flow equation of the finiteness note §4 reduces to $\partial_s B_i = \Delta B_i - \partial_i(\partial_j B_j)$, and a transverse initial field stays transverse, so

$$\hat{B}_i(t, k) = e^{-tk^2} \hat{A}_i(k), \quad \hat{b}_i(t, k) = i\varepsilon_{ijl} k_j e^{-tk^2} \hat{A}_l(k).$$

The flow time has dimension of length squared; the conventional smearing radius is $\sqrt{8t}$. Set $\varphi_t = \frac{1}{2}(b_t \cdot b_t - \langle b_t \cdot b_t \rangle_0)$, of dimension L^{-4} .

3. The susceptibility

Write $G_{ij}(z) = \langle b_{t,i}(z) b_{t,j}(0) \rangle_0$. Using $\varepsilon_{iab} \varepsilon_{jcd} k_a k_c P_{bd}^T(k) = k^2 P_{ij}^T(k)$ (the longitudinal part of P^T drops because $\varepsilon_{iab} k_a \hat{k}_b = 0$),

$$\tilde{G}_{ij}(k) = k^2 P_{ij}^T(k) \frac{g^2}{2k} e^{-2tk^2} = \frac{g^2 k}{2} e^{-2tk^2} P_{ij}^T(k).$$

For a Gaussian field Wick's theorem gives $\langle \varphi_t(z) \varphi_t(0) \rangle_0^c = \frac{1}{2} \sum_{ij} G_{ij}(z)^2$ (two pairings), so by Parseval, with $\sum_{ij} [P_{ij}^T]^2 = \text{tr } P^T = 2$,

$$\chi_{\varphi_t} = \int d^3 z \langle \varphi_t \varphi_t \rangle^c = \frac{1}{2} \int \frac{d^3 k}{(2\pi)^3} \frac{g^4 k^2}{2} e^{-4tk^2} = \frac{g^4}{4} \cdot \frac{1}{2\pi^2} \int_0^\infty dk k^4 e^{-4tk^2}.$$

With $\int_0^\infty k^4 e^{-\alpha k^2} dk = 3\sqrt{\pi}/(8\alpha^{5/2})$ and $\alpha = 4t$, so $(4t)^{5/2} = 32t^{5/2}$,

$$\chi_{\varphi_t} = \frac{g^4}{4} \cdot \frac{1}{2\pi^2} \cdot \frac{3\sqrt{\pi}}{256 t^{5/2}} = \frac{3g^4}{2048 \pi^{3/2} t^{5/2}}.$$

Dimension: $[\chi] = [\varphi]^2 L^3 = L^{-5}$ and $t^{5/2} \sim L^5$, as required.

4. The gradient term

By Parseval $O = \int d^3 y \varphi_t = \frac{1}{2} \int \frac{d^3 k}{(2\pi)^3} k^2 e^{-2tk^2} |\hat{A}(k)|^2 + \text{const}$, so

$$D_i(x) = \frac{\delta O}{\delta A_i(x)} = (-\Delta e^{2t\Delta} A)_i(x) = (\nabla \times b_{2t})_i(x), \quad \hat{D}_i(k) = k^2 e^{-2tk^2} \hat{A}_i(k),$$

the curl of the magnetic field at twice the flow time. Hence

$$\langle |D|^2 \rangle_0 = \int \frac{d^3 k}{(2\pi)^3} k^4 e^{-4tk^2} \cdot 2 \cdot \frac{g^2}{2k} = g^2 \cdot \frac{1}{2\pi^2} \int_0^\infty dk k^5 e^{-4tk^2} = \frac{g^2}{2\pi^2} \cdot \frac{1}{64 t^3} = \frac{g^2}{128 \pi^2 t^3},$$

using $\int_0^\infty k^5 e^{-\alpha k^2} dk = 1/\alpha^3$. Dimension: $[\varphi]^2 L^2 = L^{-6}$ and $t^3 \sim L^6$, as required.

5. The bound, and the cancellation of the coupling

$$\frac{\langle |D|^2 \rangle}{\chi} = \frac{g^2}{128 \pi^2 t^3} \cdot \frac{2048 \pi^{3/2} t^{5/2}}{3g^4} = \frac{16}{3g^2 \sqrt{\pi t}},$$

$$m \leq \frac{g^2 \hbar c}{2} \cdot \frac{16}{3g^2 \sqrt{\pi t}} = \frac{8}{3\sqrt{\pi}} \frac{\hbar c}{\sqrt{t}} = \frac{16\sqrt{2}}{3\sqrt{\pi}} \frac{\hbar c}{\sqrt{8t}} \approx 4.26 \frac{\hbar c}{\sqrt{8t}}.$$

Correction. Section 5 of the finiteness note inferred $m \leq C g^2 \hbar c / \sqrt{8t}$ from dimensional analysis, assigning the pure numbers k_1, k_2 no coupling dependence. The computation shows $\chi \propto g^4$ while $\langle |D|^2 \rangle \propto g^2$, so the ratio carries g^{-2} and the bound is of order g^0 . The corrected statement is that the perturbative bound is $4.26 \hbar c / \sqrt{8t} [1 + O(g^2)]$, with the coupling entering only through the corrections. The t -dependence $1/\sqrt{8t}$ of the earlier note stands.

6. Independent check: the two-photon sum rule

By the spectral reading of the finiteness note §3, the bound equals the mean excitation energy weighted by the spectral weight of O . In the free theory O is quadratic in $\hat{A}$ and creates two photons of momenta k and $-k$, of total energy $2\hbar ck$, with weight proportional to $[k^2 e^{-2tk^2}]^2 [g^2/2k]^2 k^2 dk \propto k^4 e^{-4tk^2} dk$ after the phase-space factor. Therefore

$$\frac{\int_0^\infty 2\hbar ck \cdot k^4 e^{-4tk^2} dk}{\int_0^\infty k^4 e^{-4tk^2} dk} = 2\hbar c \frac{1/(64t^3)}{3\sqrt{\pi}/(256 t^{5/2})} = 2\hbar c \cdot \frac{4}{3\sqrt{\pi t}} = \frac{8}{3\sqrt{\pi}} \frac{\hbar c}{\sqrt{t}},$$

the same number. The agreement checks Proposition 1, the intensive form of Corollary 2 and the Gaussian computation against each other, since the two routes share no step beyond the definition of φ_t .

7. What this says about the interacting theory

The free limit is correct. For $m = 0$ the bound must be able to vanish, and it does, like $1/\sqrt{8t}$: the inequality is not vacuous, and the flow time is the only resolution scale in the free theory.

Weak coupling proves nothing. In the interacting theory the perturbative regime at flow time t is $\sqrt{8t} \ll \hbar c/(\hbar c\Lambda)$, where the flowed correlators are the free ones up to $O(g^2(\sqrt{t}))$. There the bound reads $m \lesssim 4.26 \hbar c/\sqrt{8t}$, which is weaker than $\hbar c\Lambda$ by the ratio $1/(\sqrt{8t}\Lambda) \gg 1$: no constraint on the gap. The bound reaches the expected value of m only at

$$\sqrt{8t_*} \simeq 4.26 \frac{\hbar c}{m},$$

the correlation length, and there the coefficient is a nonperturbative number, since the leading behaviour of χ_{φ_t} at $\sqrt{8t} \sim \hbar c/m$ is not given by the Gaussian formula of Section 3. This sharpens the statement of the finiteness note §5: the breakdown of the $1/\sqrt{8t}$ scaling carries the content of the problem. **Proving $m < \infty$ requires the value of one susceptibility at the confinement scale, and no expansion around the free theory supplies it.**

Which susceptibility. At $\sqrt{8t} \sim \hbar c/m$, χ_{φ_t} is the zero-momentum fluctuation of the energy density smeared over a correlation length, which in a gapped theory is finite and positive, and in a theory with no finite-energy excitations would have to vanish. It is therefore the natural target for a nonperturbative lower bound, and a lattice statement of the form $\chi_{\varphi_t} \geq ct^{-5/2}$ uniform in the cutoff would close the finiteness half.

8. Consequence for STATE

The finiteness half is now quantitative: an exact inequality, a closed free-field value with an independent check, and an explicit statement of where it becomes informative. The next step it names is a lower bound on χ_{φ_t} at $\sqrt{8t}$ of order the correlation length, uniform in the lattice spacing, which is a positivity statement about one flowed correlator rather than a spectral statement. Together with hypothesis (F1) it would give $m < \infty$; the clause $m > 0$ remains untouched by this whole line, and its obstruction is the one recorded in the Schur note.